Independence of the Lubin–Tate parameter and the local Artin action

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Source: *Class Field Theory*, v4.03, 6 August 2020

Private reading transcription of the author's Theorem I.3.9 and its supporting argument. The original PDF is retained at https://www.jmilne.org/math/CourseNotes/CFT.pdf. The main argument is on physical PDF pages 49–52 (printed pages 40–43). Context below comes from physical pages 41–42, 44–45 and 47. Original numbering is prefixed by the chapter to disambiguate it. Mathematical expressions and the author's abbreviated steps are retained; connective prose and layout are normalized. This is published-material transcription, not author-native TeX or a new proof. Author errata are identified separately below.

Author context and notation

Let K be a nonarchimedean local field, $A = \mathcal{O}_K$, with maximal ideal $\mathfrak{m}$, residue field $k = A/\mathfrak{m}$ of order q, and prime element π. Write K^{al} for the separable algebraic closure used in the source and K^{un} for its maximal unramified subextension. The source's class $\mathcal{F}_\pi$ consists of series $f \in A[[T]]$ satisfying

$$f(T) = \pi T + \text{terms of degree } \geq 2, \qquad f(T) \equiv T^q \pmod{\pi}.$$

Summary I.2.20 provides the unique formal group law F_f admitting f as an endomorphism and its unique A-module structure $a \mapsto [a]_f$, with

$$[a]_f = aT + \text{terms of degree } \geq 2, \qquad [a]_f \circ f = f \circ [a]_f, \qquad [\pi]_f = f.$$

For $g \in \mathcal{F}_\pi$, F_f and F_g have the canonical A-isomorphism denoted $[1]_{g,f} : F_f \longrightarrow F_g$ in the source.

The setup in Section I.3 is

$$\Lambda_f = \{\alpha \in K^{\mathrm{al}} : |\alpha| < 1\}, \quad \alpha +_{\Lambda_f} \beta = F_f(\alpha, \beta), \quad a * \alpha = [a]_f(\alpha).$$

Let Λ_n be the submodule annihilated by $[\pi]_f^n$; equivalently, its elements are the roots of $f^{(n)}$ of absolute value less than one. Set $K_{\pi,n} = K[\Lambda_n]$ and $K_\pi = \bigcup_n K_{\pi,n}$. The relevant preceding conclusions of I.3.4 and I.3.6 are

$$\Lambda_n \simeq A/(\pi^n), \quad [K_{\pi,n} : K] = (q-1)q^{n-1}, \quad (A/\mathfrak{m}^n)^\times \xrightarrow{\sim} \mathrm{Gal}(K_{\pi,n}/K),$$

where the last map is induced by the A-action, and $K_{\pi,n}/K$ is totally ramified. These preceding Lubin–Tate structure results are context, not additional counted targets. The author's separate errata include the convergence convention for Λ_f; the original setup is preserved here and that note remains attached to the source package.

The author's intertwining lemma

Lemma I.2.11 (support). *Let $f, g \in \mathcal{F}_\pi$, and let $\phi_1(X_1, \ldots, X_n)$ be a linear form with coefficients in A. There is a unique $\phi \in A[[X_1, \ldots, X_n]]$ such that*

$$\begin{cases} \phi(X_1, \ldots, X_n) = \phi_1 + \textit{terms of degree } \geq 2, \\ f(\phi(X_1, \ldots, X_n)) = \phi(g(X_1), \ldots, g(X_n)). \end{cases}$$

Author proof. Inductively construct the unique polynomial ϕ_r of degree r with

$$\begin{cases} \phi_r(X_1, \ldots, X_n) = \phi_1 + \text{terms of degree } \geq 2, \\ f(\phi_r(X_1, \ldots, X_n)) = \phi_r(g(X_1), \ldots, g(X_n)) + \text{terms of degree } \geq r+1. \end{cases}$$

For $r = 1$ the candidate is ϕ_1. If $\phi_1 = \sum a_i X_i$, its second condition is $\pi(\sum a_i X_i) = \sum a_i(\pi X_i) +$ terms of degree ≥ 2. After ϕ_r has been constructed, write $\phi_{r+1} = \phi_r + Q$, with Q homogeneous of degree $r+1$. The two sides of the required relation are

$$\begin{aligned} f(\phi_{r+1}(X_1, \ldots, X_n)) &= f(\phi_r(X_1, \ldots, X_n)) + \pi Q(X_1, \ldots, X_n) + \text{terms of degree } \geq r+2, \\ \phi_{r+1}(g(X_1), \ldots, g(X_n)) &= \phi_r(g(X_1), \ldots, g(X_n)) + Q(\pi X_1, \ldots, \pi X_n) + \text{terms of degree } \geq r+2. \end{aligned}$$

The source's homogeneity line prints $Q(\pi X_1, \ldots) = \pi^{r+1} Q(\pi X_1, \ldots)$; the author's erratum for printed p.32 deletes the third π, giving $\pi^{r+1} Q(X_1, \ldots)$. The following displayed requirement in the source is

$$(\pi^{r+1} - \pi) Q(X_1, \ldots, X_n) = f(\phi_r(X_1, \ldots, X_n)) - \phi_r(g(X_1), \ldots, g(X_n)) + \text{terms of degree } \geq r+2.$$

Thus Q is determined by

$$\frac{f(\phi_r(X_1, \ldots, X_n)) - \phi_r(g(X_1), \ldots, g(X_n))}{(\pi^r - 1)\pi} = Q + \text{terms of degree } \geq r+2.$$

The characteristic-p binomial calculation gives

$$f \circ \phi_r - \phi_r \circ g \equiv \phi_r(X_1, \ldots, X_n)^q - \phi_r(X_1^q, \ldots, X_n^q) \equiv 0 \pmod{\pi}.$$

The numerator is divisible by π, and $\pi^r - 1$ is a unit of A. Consequently Q has coefficients in A and, by the induction hypothesis, has degree $r+1$. The compatible polynomials determine a unique series with

$$\phi_{r+1} = \phi_r + \text{terms of degree } \geq r+1, \qquad \phi = \phi_r + \text{terms of degree } \geq r+1 \quad \text{for every } r.$$

The source ends the verification with

$$\begin{aligned} f(\phi(X_1, \ldots, X_n)) &= f(\phi_r(X_1, \ldots, X_n)) + \text{terms of degree } \geq r+1 \\ &= \phi_r(g(X_1), \ldots, g(X_n)) + \text{terms of degree } \geq r+1 \\ &= \phi(f(X_1), \ldots, f(X_n)) + \text{terms of degree } \geq r+1. \end{aligned}$$

The last line is transcribed with f as printed in the original PDF. The author concludes that, as this holds for all r, the second required property holds. □

The local Artin map and the target statement

Define

$$\phi_\pi : K^\times \longrightarrow \mathrm{Gal}((K_\pi \cdot K^{\mathrm{un}})/K).$$

Because $K_\pi \cap K^{\mathrm{un}} = K$, the two restrictions specify its action. For $a = u\pi^m$, $u \in A^\times$, its restriction to K^{un} is Frob_K^m, and its action on the torsion generators of K_π is

$$\phi_\pi(a)(\lambda) = [u^{-1}]_f(\lambda), \qquad \lambda \in \bigcup_n \Lambda_n.$$

Theorem I.3.9. *Both $K_\pi \cdot K^{\mathrm{un}}$ and ϕ_π are independent of the choice of π.*

Recall that K^{un} is not complete. Let $\widehat{K^{\mathrm{un}}}$ be its completion, and let B be its valuation ring. Write σ for Frob_K and its extension to $\widehat{K^{\mathrm{un}}}$. For $\theta(T) = \sum b_i T^i \in B[[T]]$, set $(\sigma\theta)(T) = \sum(\sigma b_i)T^i$.

Proposition I.3.10 (support). *Let $f \in \mathcal{F}_\pi$ and $g \in \mathcal{F}_\varpi$, where $\varpi = u\pi$. The formal group laws F_f, F_g become A-isomorphic over B. More precisely, there are $\varepsilon \in B^\times$ and $\theta(T) \in B[[T]]$ with $\sigma\varepsilon = \varepsilon u$ such that*

(a) $\theta(T) = \varepsilon T +$ *terms of degree* ≥ 2,
(b) $\sigma\theta = \theta \circ [u]_f$,
(c) $\theta(F_f(X,Y)) = F_g(\theta(X), \theta(Y))$,
(d) $\theta \circ [a]_f = [a]_g \circ \theta$.

Conditions (c),(d) express compatibility with the formal groups and A-actions; (a) makes the homomorphism an isomorphism, since ε is a unit.

Lemma I.3.11 (support). *The maps*

$$b \longmapsto \sigma b - b : B \longrightarrow B, \qquad b \longmapsto \frac{\sigma b}{b} : B^\times \longrightarrow B^\times$$

are surjective, with kernels A and $A^\times$, respectively.

Author proof. Let R be the valuation ring of K^{un} and $\mathfrak{n}$ its maximal ideal. Then R is a discrete valuation ring and $\varprojlim R/\mathfrak{n}^n = B$ (I.A.7). The author proves inductively that

$$0 \longrightarrow A/\mathfrak{m}_K^n \longrightarrow R/\mathfrak{n}^n \xrightarrow{\sigma-1} R/\mathfrak{n}^n \longrightarrow 0 \tag{10}$$

is exact. At $n = 1$ it becomes

$$0 \longrightarrow k \longrightarrow \bar{k} \xrightarrow{x \mapsto x^q - x} \bar{k} \longrightarrow 0,$$

where $\bar{k}$ is the algebraic closure of k. For the induction step, use

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R/\mathfrak{n} & \longrightarrow & R/\mathfrak{n}^n & \longrightarrow & R/\mathfrak{n}^{n-1} & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \sigma-1} & & \downarrow{\scriptstyle \sigma-1} & & \downarrow{\scriptstyle \sigma-1} & & \\ 0 & \longrightarrow & R/\mathfrak{n} & \longrightarrow & R/\mathfrak{n}^n & \longrightarrow & R/\mathfrak{n}^{n-1} & \longrightarrow & 0. \end{array}$$

The snake lemma gives surjectivity on $R/\mathfrak{n}^n$ and a kernel with q^n elements. The image of $A/\mathfrak{n}^n$, already contained in that kernel, has that size. Passing to the inverse limit yields the exact sequence

$$0 \longrightarrow A \longrightarrow B \xrightarrow{\sigma-1} B \longrightarrow 0$$

by I.A.8. The author states that the proof for $B^\times$ is similar. □

Author proof of Proposition I.3.10

Composition inverses of series are written h^{-1}, so $h \circ h^{-1} = T = h^{-1} \circ h$. The author divides the proof into four steps:

1. Construct $\theta \in B[[T]]$ satisfying (a),(b).
2. Choose it so that $g = \sigma\theta \circ f \circ \theta^{-1}$.
3. Show that $\theta(F_f(\theta^{-1}(X), \theta^{-1}(Y)))$ has the properties characterizing $F_f(X, Y)$, hence equals it.[1]
4. Show that $\theta \circ [a]_f \circ \theta^{-1}$ has the properties characterizing $[a]_g$, hence equals it.

Step 1

Choose $\varepsilon \in B^\times$ with $\sigma\varepsilon = \varepsilon u$ using I.3.11. Starting from $\theta_1(T) = \varepsilon T$, construct polynomials with

$$\theta_r(T) = \theta_{r-1}(T) + bT^r \quad (b \in B), \qquad \sigma\theta_r = \theta_r \circ [u]_f + \text{terms of degree } \geq r+1.$$

The first case is $\sigma\varepsilon T = \varepsilon(uT + \cdots) + \text{terms of degree } \geq 2$. Given θ_r, set $\theta_{r+1} = \theta_r + bT^{r+1}$ and write $b = a\varepsilon^{r+1}$. The desired relation is

$$(\sigma\theta_r)(T) + (\sigma a)(\sigma\varepsilon)^{r+1}T^{r+1} = \theta_r([u]_f(T)) + a\varepsilon^{r+1}(uT)^{r+1} + \text{terms of degree } \geq r+1.$$

Thus the required coefficient equation is

$$(\sigma a - a)(\varepsilon u)^{r+1} = c,$$

where c is the coefficient of T^{r+1} in $\theta_r \circ [u]_f - \sigma\theta_r$. Choose $a \in B$ with $\sigma a - a = c/(\varepsilon u)^{r+1}$.

Step 2

Define

$$h = \sigma\theta \circ f \circ \theta^{-1} = \theta \circ [u]_f \circ f \circ \theta^{-1} = \theta \circ f \circ [u]_f \circ \theta^{-1}.$$

Since f and $[u]_f$ have coefficients in A,

$$\sigma h = \sigma\theta \circ f \circ [u]_f \circ \sigma\theta^{-1} = \sigma\theta \circ f \circ \theta^{-1} = h.$$

Here $[u]_f \circ \sigma\theta^{-1} = \theta^{-1}$ follows from $\theta \circ [u]_f \circ \sigma\theta^{-1} = T$. It follows that $h \in A[[T]]$. Moreover,

$$h(T) = \sigma\varepsilon \cdot \pi \cdot \varepsilon^{-1}T + \cdots = \varpi T + \text{terms of degree } \geq 2,$$

and

$$h(T) \equiv \sigma\theta \circ (\theta^{-1})^q = \sigma\theta(\sigma\theta^{-1}(T^q)) \equiv T^q \pmod{\mathfrak{m}_K}.$$

Thus $h \in \mathcal{F}_\varpi$. Set $\theta' = [1]_{g,h} \circ \theta$. It continues to satisfy (a),(b), because $[1]_{g,h} \in A[[T]]$, and

$$\sigma\theta' \circ f \circ (\theta')^{-1} = [1]_{g,h} \circ h \circ [1]_{g,h}^{-1} = g.$$

The author leaves Steps 3 and 4 as straightforward applications of I.2.11.

[1] The original Step 3 says F_f; the author's separate erratum for printed p.41 specifies F_g here. This source correction is identified rather than silently applied.

Author proof that $K_\pi K^{\mathrm{un}}$ is independent of π

Let $\varpi = u\pi$. Proposition I.3.10 gives

$$(\sigma\theta) \circ [\pi]_f = \theta \circ [u]_f \circ [\pi]_f = \theta \circ [\varpi]_f = [\varpi]_g \circ \theta,$$

or $(\sigma\theta)(f(T)) = g(\theta(T))$. For $\alpha \in K^{\mathrm{al}}$,

$$f(\alpha) = 0 \implies g(\theta(\alpha)) = 0, \qquad g(\alpha) = 0 \implies f(\theta^{-1}(\alpha)) = 0.$$

Hence θ gives a bijection $\Lambda_{f,1} \longrightarrow \Lambda_{g,1}$, and

$$\begin{aligned}\widehat{K}^{\mathrm{un}}[\Lambda_{g,1}] &= \widehat{K}^{\mathrm{un}}[\theta(\Lambda_{f,1})] \subset \widehat{K}^{\mathrm{un}}[\Lambda_{f,1}] \\ &= \widehat{K}^{\mathrm{un}}[\theta^{-1}(\Lambda_{g,1})] \subset \widehat{K}^{\mathrm{un}}[\Lambda_{g,1}].\end{aligned}$$

Thus $\widehat{K}^{\mathrm{un}}[\Lambda_{g,1}] = \widehat{K}^{\mathrm{un}}[\Lambda_{f,1}]$. The next lemma provides

$$\widehat{K}^{\mathrm{un}}[\Lambda_{g,1}] \cap K^{\mathrm{al}} = K^{\mathrm{un}}[\Lambda_{g,1}], \qquad \widehat{K}^{\mathrm{un}}[\Lambda_{f,1}] \cap K^{\mathrm{al}} = K^{\mathrm{un}}[\Lambda_{f,1}],$$

so $K^{\mathrm{un}}[\Lambda_{g,1}] = K^{\mathrm{un}}[\Lambda_{f,1}]$. The author extends the same argument to every n, obtaining

$$K^{\mathrm{un}}[\Lambda_{g,n}] = K^{\mathrm{un}}[\Lambda_{f,n}], \qquad K^{\mathrm{un}} \cdot K_\varpi = K^{\mathrm{un}} \cdot K_\pi.$$

Lemma I.3.12 (support). *Every subfield E of K^{al} containing K is closed in the topological sense.*

Author proof. Put $H = \mathrm{Gal}(K^{\mathrm{al}}/E)$. It fixes E and therefore, by continuity, every element in the closure of E. Galois theory then identifies E with its closure in K^{al}. □

Author proof that ϕ_π is independent of π

For two prime elements π, ϖ, the author proves $\phi_\pi(\varpi) = \phi_\varpi(\varpi)$. Since π is arbitrary, another prime element π' then satisfies

$$\phi_{\pi'}(\varpi) = \phi_\varpi(\varpi) = \phi_\pi(\varpi).$$

Prime elements generate $K^\times$, so this proves $\phi_\pi = \phi_{\pi'}$. On K^{un} the two maps both induce Frobenius; compare them on K_ϖ. The isomorphism $\theta : F_f \to F_g$ of I.3.10 induces $\Lambda_{f,n} \to \Lambda_{g,n}$. By definition $\phi_\varpi(\varpi)$ is the identity on K_ϖ. Since the elements $\theta(\lambda)$ for $\lambda \in \Lambda_{f,n}$ generate $K_{\varpi,n}$, it remains to show

$$\phi_\pi(\varpi)(\theta(\lambda)) = \theta(\lambda), \qquad \lambda \in \Lambda_{f,n}.$$

Write $\varpi = u\pi$ and $\phi_\pi(\varpi) = \phi_\pi(u)\phi_\pi(\pi) = \tau\sigma$, where

$$\sigma = \begin{cases} \mathrm{Frob}_K & \text{on } K^{\mathrm{un}}, \\ \mathrm{id} & \text{on } \lambda, \end{cases} \qquad \tau = \begin{cases} \mathrm{id} & \text{on } K^{\mathrm{un}}, \\ [u^{-1}]_f & \text{on } \lambda. \end{cases}$$

Using the coefficients of θ in $\widehat{K}^{\mathrm{un}}$ and I.3.10, the author's final identity is

$$\phi_\pi(\varpi)(\theta(\lambda)) = \tau\sigma(\theta(\lambda)) = (\sigma\theta)(\tau\lambda) = (\sigma\theta)([u^{-1}]_f(\lambda)) = \theta(\lambda).$$

Source and revision map

The primary statement, comparison proposition, difference lemma and Artin-map definition are on physical p.49 / printed p.40. The difference-lemma proof and start of Step 1 are on p.50 / p.41; Step 1, Step 2 and the first field-comparison argument continue on p.51 / p.42. The completion-to-algebraic comparison, closed-field lemma and final Artin comparison are on p.52 / p.43. Lemma I.2.11 is on p.41–42 / p.32–33. The formal-module summary is p.44 / p.35; torsion definitions are p.45 / p.36; the structure results are p.47 / p.38. The complete author PDF and the author's separately saved errata remain the authoritative sources. No support result in this reader is counted as another corpus problem.